

Reach $\neq$ equity: the plans that serve the most sites reach only a quarter of demand in the neediest third of neighborhoods.

Motivation and Approach

Urban food rescue differs fundamentally from commercial delivery: nonprofit organizations must recover uncertain donations, preserve perishable food, meet donor and recipient time windows, and decide which communities to serve when capacity cannot reach them all. **Once service becomes optional, every route decides which neighborhoods go without that day.**

We formulate the problem as an extension of the vehicle routing problem with time windows, incorporating donor pickups, recipient opening hours, vehicle and refrigerated capacity, perishability penalties, and an equity-weighted value for serving each recipient. Prior work covers parts of this problem, from time-windowed routing [5] to humanitarian pickup and distribution [6] and equitable food allocation [7], but none places a spatial measure of access inside the routing objective. Using public New York City data on 528 food pantries and kitchens, and grounded in conversations with nonprofit food-rescue staff and drivers, the framework compares objectives that define equity differently. The central question is how much breadth and travel must be traded to direct scarce food toward the neighborhoods that need it most.

2.2M

New Yorkers in poverty
in 2024, a record [1]

62%

of SNAP families ran out
of food, or feared it [1]

5–35%

food-insecurity rate,
varying by neighborhood [2]

Research Questions

- How does adding neighborhood need and spatial access to the routing objective change who gets served, and does the answer depend on how equity is measured?
- What does it cost in breadth (number of sites reached) and travel (fleet vehicle-minutes), and how hard should equity be weighted before the gains in coverage flatten out?

Building the Equity Map

Asking who is underserved requires two measurements that do not overlap. **Need** N_i is the neighborhood's food-insecurity rate, a modeled estimate from the Mayor's Office of Food Policy and HRA [2]. It is preferred to SNAP enrollment, which counts who has found help rather than who needs it. **Access** A_i is a Gaussian-decay E2SFCA score [3] over 2,325 census tracts and 526 food providers, crediting each provider's weekly open hours S_j to every tract within a one-mile walk, discounted by distance so that nearby providers count the most:

$$R_j = \frac{S_j}{\sum_i D_i W(d_{ij})}, \quad A_i = \sum_j R_j W(d_{ij})$$

Tract weights D_i are total population, not food-insecure population, so the two scores stay independent by construction; both are then rescaled to percentile ranks from 0 to 1.

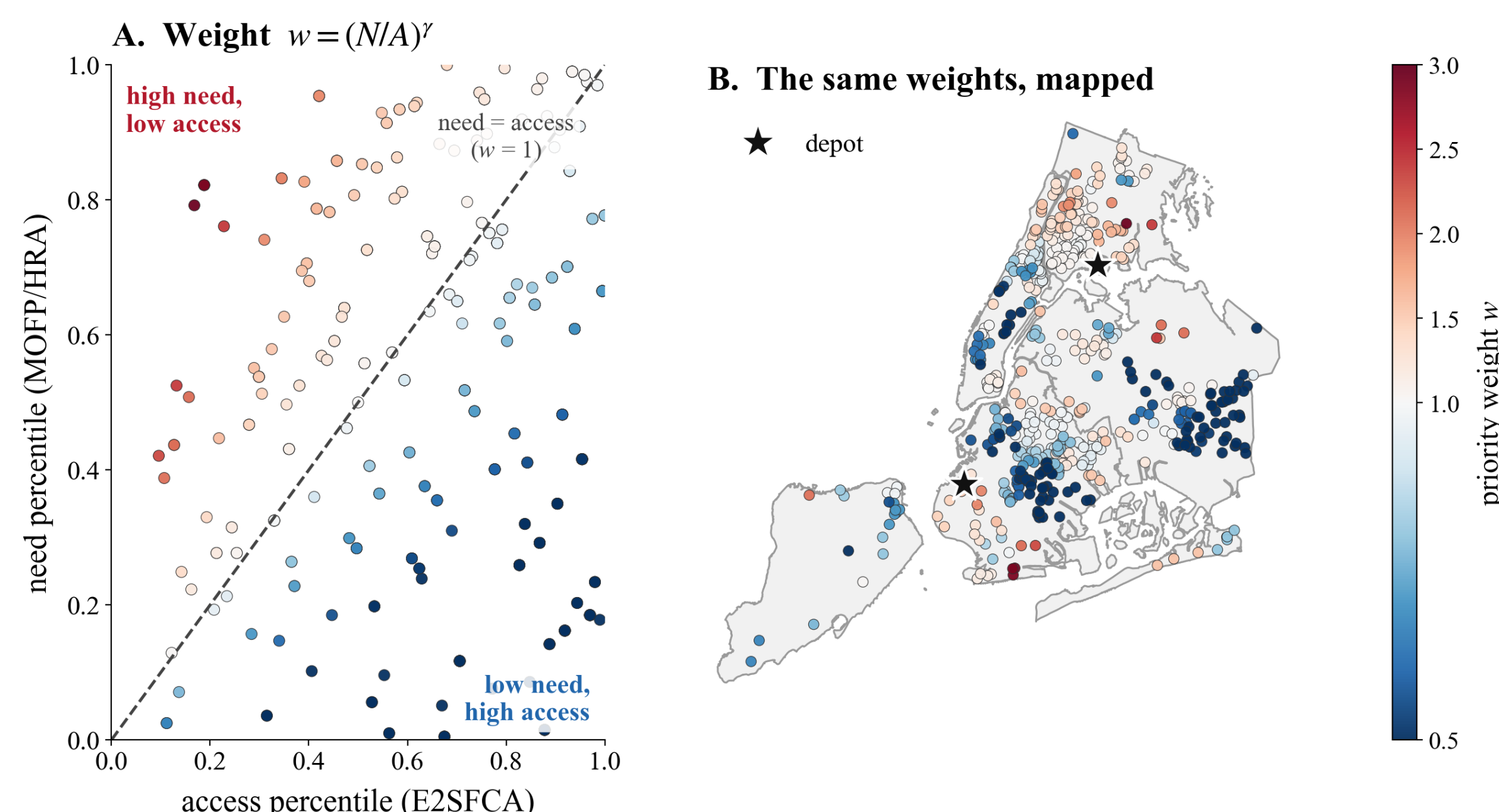


Figure 1. Priority weights at $\gamma = 1$: by need and access percentile (A) and on the map (B).

The Model

An optional-service pickup-and-delivery VRPTW. Binary y_i marks whether recipient $i \in V$ is served, and x_{ijk} whether vehicle $k \in K$ drives from i to j . Each recipient's priority weight is

$$w_i = \text{clip}\left(\left(\frac{N_i + \varepsilon}{A_i + \varepsilon}\right)^\gamma, w_{\min}, w_{\max}\right),$$

where ε stabilizes the ratio near zero access. The exponent γ is an **equity dial**: at $\gamma = 0$ every recipient counts equally; larger γ favors underserved areas. The solver maximizes

$$\sum_{i \in V} P w_i y_i - \lambda \sum_{k \in K} \sum_{(i,j)} c_{ij} x_{ijk} - \delta \sum_{i \in V} \text{cold}_i \tau_i,$$

where P is the base value of serving a recipient, c_{ij} the travel time from i to j , cold_i the refrigerated pounds due at i , and τ_i the arrival time there; λ and δ set how heavily travel and late cold food count, subject to capacity, refrigeration, and time windows. Because y_i is a decision rather than a requirement, the model must choose whom to serve. **This is when the allocation comes in.**

Example instance: 528 public recipient locations [4], 18 donors, 25 vehicles, 2 depots, and OSRM road travel times. Fleet and demand parameters are labeled scenario assumptions.

Solution method: routes are optimized with Google OR-Tools using guided local search, under a 30-second time budget per solve. Because a single solve varies by a few points, each strategy is solved five times with small random perturbations and reported by its median.

Strategies Compared

Every strategy faces the same 25 trucks, the same donated food, and the same opening hours, so any difference in who gets served traces back to the objective function alone.

Strategy	Priority weight	Equity signal
Unweighted	1 ($\gamma = 0$)	none
Random-preference	$\mathcal{U}(0.3, 1.7)$	arbitrary
Need-only	need term only	need
Access-only	access term only	access
Need-access	$(N_i/A_i)^\gamma, \gamma = 1$	need $\div$ access

Results Across Strategies

Table 1. Medians of five replicate solves, need-access at $\gamma = 1$; highlights exceed replicate noise.

Metric	Unwt.	Random	Need	Access	Need-acc.
Sites served	231	234	209	194	183
High-need tier %	26	25	58	22	44
Need-weighted % [†]	29	31	36	42	46
Travel (veh-min)	12,412	13,034	12,991	13,501	13,146

[†]Weights each recipient by w at $\gamma = 1$: the ratio need-access maximizes.

Without an equity signal, service ignores need. Unweighted and Random serve the most sites, yet reach only 26% and 25% of the *high-need tier*: demand served among recipients in the neediest third of neighborhoods, a measure built from neighborhood need alone.

The yardstick picks the winner. Need-only reaches 58% of that tier and still serves more sites than need-access does. Need-access leads only on need-weighted coverage, which scores each recipient by the very ratio it optimizes. By that same measure Access-only ranks second, at 42%, while doing no better than the two no-signal plans do on the high-need tier.

Travel barely moves. Median fleet time differs by at most 9% across all five strategies, comparable to the 7 to 14% that a single strategy varies between its replicate solves.

The Equity Dial

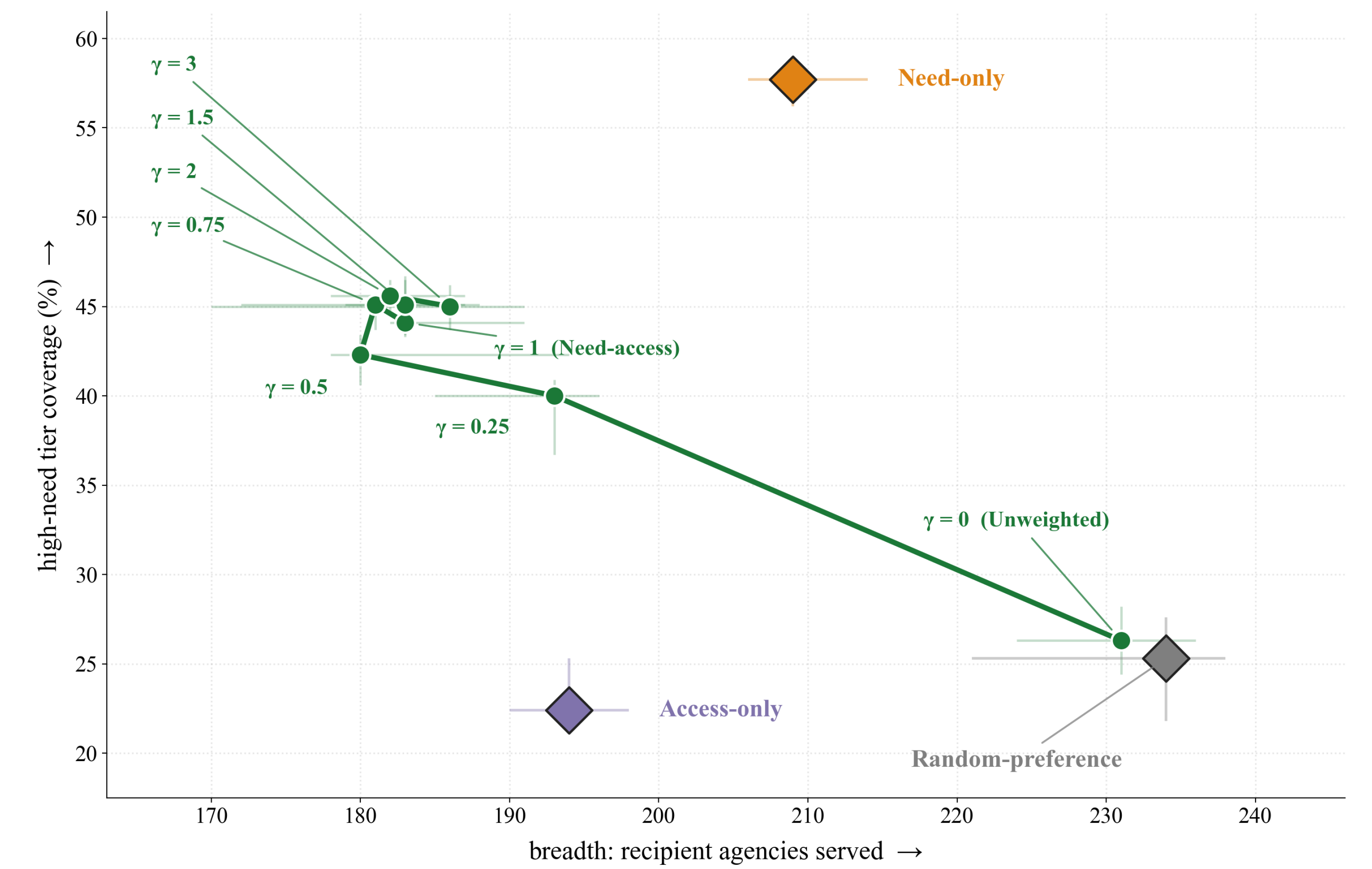


Figure 2. Every strategy on two axes no objective optimizes directly. Points are medians of five replicate solves; whiskers show the range. The green curve is need-access as γ rises; its $\gamma = 0$ end is Unweighted.

Most of the gain, and most of the cost, arrive in the first step: $\gamma = 0.25$ lifts high-need coverage from 26% to 40% and cuts sites served from 231 to 193. Past $\gamma \approx 0.75$ both plateau, near 45% and 183 sites, and the remaining wiggles sit inside replicate noise. **Need-only lies above and to the right of the entire curve**; even its worst replicate beats the curve's best on both axes. No setting of the dial reaches it; folding in access dilutes the model's focus on need.

Conclusion and Broader Impact

- Routing objectives are allocation policies.** Under scarcity every plan leaves someone unserved; without an equity signal, the choice falls to travel cost or to chance.
- A fairness metric can grade itself.** Scored by its own weights, need-access looks like the best choice. Scored by need alone, need-only wins on both coverage and breadth, by margins wider than replicate noise, so the choice of yardstick is itself a policy decision.
- Equity costs breadth, not driving.** Need-access served about a fifth fewer sites than Unweighted (183 versus 231), while total fleet time stayed within replicate noise.

This is decision-support evidence, not an operational recommendation: need is measured by neighborhood, access by provider hours, and the model plans a single representative day. Recipient-capacity uncertainty and delivery recourse are the next step to evaluate.

Broader impact. Nonprofit staff described operational capacity, not only food supply, as what often limits how many people they reach. Allocation is therefore a lever that requires no additional trucks: the same fleet can be directed toward the neighborhoods that need it most, with no measurable change in driving. The framework also makes an otherwise invisible choice auditable, showing planners before dispatch which communities a plan leaves out. The same logic applies wherever demand exceeds capacity, from disaster relief to mobile health clinics and emergency supply delivery, wherever a fixed fleet must decide whom to reach first.

References

- [1] Robin Hood & Columbia Center on Poverty and Social Policy, *Poverty Tracker Annual Report*, Vol. 8, 2026.
- [2] NYC Council Data Team, "Emergency Food in NYC"; NYC MOFP/HRA neighborhood prioritization data, 2026.
- [3] Luo & Qi, *Health & Place* 15(4), 2009.
- [4] NYC HRA, Food Help NYC locations, 2026.
- [5] Solomon, *Operations Research* 35(2), 1987.
- [6] Eisenhandler & Tzur, *Operations Research* 67(1), 2019.
- [7] Orgut, Ivy, Uzsoy & Wilson, *IIE Transactions* 48(3), 2016.